

Categorical Data Analysis

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1 Problem 1: Lung Cancer Data Analysis

1.1 Chi-Square Test Interpretation

The Chi-Square test produced a statistic value of:

$$\chi^2 = 67.59$$

with a p-value of:

$$p = 2.01 \times 10^{-16}$$

Since the p-value is much smaller than the significance level

$$\alpha = 0.05,$$

the null hypothesis is rejected. Therefore, a statistically significant association exists between the variables under study.

1.2 Pollution vs Lung Disease

For the relationship between pollution and lung disease, the Chi-Square test yielded:

$$\chi^2 = 33.84$$

and

$$p = 5.98 \times 10^{-9}$$

Since the p-value is less than 0.05, the null hypothesis is rejected. Therefore, pollution exposure has a statistically significant association with lung disease.

Individuals exposed to higher levels of pollution are more likely to suffer from lung disease.

1.3 Fisher's Exact Test Interpretation

Fisher's Exact Test was conducted because:

- the sample size is small,
- expected cell counts are low,
- and the contingency table is 2×2 .

The test results are:

$$\text{Odds Ratio} = 5.03$$

$$p = 5.13 \times 10^{-17}$$

Since the p-value is significantly smaller than 0.05, the null hypothesis is rejected.

This indicates a statistically significant relationship between smoking and disease occurrence.

The odds ratio suggests that smokers have approximately five-fold higher odds of developing lung disease compared to non-smokers.

1.4 Correlation Matrix Interpretation

The correlation matrix between ID and Age is shown below:

	ID	Age
ID	1.000	0.023
Age	0.023	1.000

The correlation coefficient between ID and Age is:

$$r = 0.023$$

This value indicates an extremely weak positive correlation between ID and Age.

Since the coefficient is very close to zero, there is practically no linear relationship between these variables.

2 Problem 2: One-Way ANOVA

2.1 ANOVA Test Interpretation

The ANOVA test produced an F-statistic of:

$$F = 30.81$$

with a p-value of:

$$p = 7.65 \times 10^{-11}$$

Since the p-value is far smaller than the significance level

$$\alpha = 0.05,$$

the null hypothesis is rejected.

This indicates that there is a statistically significant difference among the group means.

At least one group mean differs significantly from the others.

2.2 Levene's Test Interpretation

Levene's Test was conducted to check the assumption of homogeneity of variances.

The results are:

$$\text{Levene Statistic} = 2.18$$

$$p = 0.119$$

Since the p-value is greater than 0.05, the null hypothesis of equal variances cannot be rejected.

Therefore, the assumption of homogeneity of variance is satisfied and the ANOVA results are reliable.

3 Problem 3: Chi-Square Test on Wealth Index and Anemia

A Chi-Square test of independence was conducted to examine the association between wealth index and anemia level.

3.1 Contingency Table

Wealth Index	Mild	Moderate	Not Anemic	Severe
Middle	596	865	729	59
Poorer	525	926	498	87
Poorest	522	983	418	119
Richer	606	750	737	48
Richest	505	403	797	9

3.2 Chi-Square Test Results

$$\chi^2 = 530.40$$

$$df = 12$$

$$p = 7.45 \times 10^{-106}$$

Since the p-value is far smaller than 0.05, the null hypothesis is rejected.

Therefore, there is a highly statistically significant association between wealth index and anemia level.

The contingency table suggests that poorer and poorest individuals experience higher frequencies of moderate and severe anemia, whereas the richest group contains a higher proportion of non-anemic individuals.

These findings indicate that socioeconomic status plays an important role in anemia prevalence.

4 Problem 4: Fisher's Exact Test for Drug Treatments

Fisher's Exact Test was conducted to determine whether drug treatment is associated with disease outcome.

4.1 Contingency Table

Treatment	No Disease	Disease
Drug A	40	10
Drug B	10	40
Drug C	25	25

4.2 Fisher's Exact Test Results

$$p = 1.52 \times 10^{-8}$$

Since the p-value is significantly smaller than 0.05, the null hypothesis is rejected.

Therefore, there is a statistically significant association between drug treatments and disease outcomes.

Drug A demonstrated comparatively higher treatment effectiveness because it produced the highest number of non-diseased individuals.

Drug B demonstrated comparatively lower treatment effectiveness because it produced the highest number of diseased individuals.

Drug C showed moderate treatment effectiveness.

5 Problem 5: Logistic Regression Interpretation

A logistic generalized linear model (GLM) was fitted to examine how GRE score, GPA, and undergraduate institution rank affect graduate school admission.

5.1 Overall Model Significance

The likelihood ratio test produced:

$$p = 8.207 \times 10^{-9}$$

Since the p-value is much smaller than 0.05, the logistic regression model is statistically significant.

The pseudo R^2 value is:

$$R^2 = 0.081$$

This indicates that approximately 8.1% of the variation in admission outcome is explained by the predictors.

5.2 GRE Score

$$\beta_{\text{GRE}} = 0.0023$$

$$p = 0.036$$

Since the p-value is less than 0.05, GRE score significantly affects admission probability.

Higher GRE scores increase the likelihood of graduate admission.

5.3 GPA

$$\beta_{\text{GPA}} = 0.7770$$

$$p = 0.018$$

Since the p-value is below 0.05, GPA significantly influences graduate admission. Students with higher GPAs are more likely to gain admission.

5.4 Institution Rank

$$\beta_{\text{Rank}} = -0.5600$$

$$p < 0.001$$

The negative coefficient indicates that higher-ranked institutions increase admission probability.

Institution rank demonstrated a strong statistical effect on admission decisions.

6 Problem 6: Poisson Regression Interpretation

A Poisson generalized linear model (GLM) was fitted to identify factors associated with the number of awards earned by students.

6.1 Model Assumptions

The assumptions of the Poisson regression model include:

- independence of observations,
- count nature of the response variable,
- equality of mean and variance.

6.2 Math Score

$$\beta_{\text{math}} = 0.0861$$

$$p < 0.001$$

Since the p-value is statistically significant, math score significantly affects the number of awards.

The incidence rate ratio (IRR) is:

$$IRR = e^{0.0861} \approx 1.09$$

This means that each one-point increase in math score increases the expected number of awards by approximately 9%.

6.3 Program Type

$$\beta_{\text{prog}} = 0.1233$$

$$p = 0.450$$

Since the p-value is greater than 0.05, program type is not statistically significant.

7 Problem 7: Negative Binomial Regression Interpretation

A Negative Binomial generalized linear model was fitted to identify factors associated with student absenteeism.

7.1 Math Score

$$\beta_{\text{math}} = -0.0067$$

$$p = 0.007$$

Higher math scores are associated with lower absenteeism.

$$IRR = e^{-0.0067} \approx 0.993$$

Each one-point increase in math score decreases expected absence days by approximately 0.7%.

7.2 Program Type

$$\beta_{\text{prog}} = -0.6781$$

$$p < 0.001$$

Program type significantly affects absenteeism.

$$IRR = e^{-0.6781} \approx 0.51$$

Students in stronger academic programs have approximately 49% fewer absence days.

8 Problem 8: Zero-Inflated Poisson Regression Interpretation

A Zero-Inflated Poisson (ZIP) regression model was fitted to identify factors associated with the number of fish caught.

8.1 Count Model

8.1.1 Children

$$\beta_{\text{child}} = -1.1367$$

$$IRR = e^{-1.1367} \approx 0.32$$

Groups with more children catch fewer fish.

8.1.2 Persons

$$\beta_{\text{persons}} = 0.8290$$

$$IRR = e^{0.8290} \approx 2.29$$

Larger groups catch more fish.

8.1.3 Camper

$$\beta_{\text{camper}} = 0.7243$$

$$IRR = e^{0.7243} \approx 2.06$$

Campers tend to catch more fish than non-campers.

9 Problem 9: Zero-Inflated Negative Binomial Regression Interpretation

A Zero-Inflated Negative Binomial (ZINB) regression model was fitted to account for excess zeros and overdispersion in fish count data.

9.1 Dispersion Parameter

$$\alpha = 2.0825$$

The large positive dispersion parameter confirms substantial overdispersion in the data.

9.2 Model Warning

The model failed to converge completely:

$$\text{converged} = \text{False}$$

Therefore, the coefficient estimates should be interpreted cautiously.

10 Problem 10: Zero-Truncated Poisson Regression Interpretation

A Zero-Truncated Poisson regression model was fitted to identify factors associated with hospital stay duration.

10.1 Age

$$\beta_{\text{age}} = -0.0144$$

$$IRR = e^{-0.0144} \approx 0.986$$

Older patients showed slightly shorter hospital stays.

10.2 Health Insurance

$$\beta_{\text{hmo}} = -0.1359$$

$$IRR = e^{-0.1359} \approx 0.873$$

Patients with HMO insurance had shorter hospital stays.

11 Problem 11: Zero-Truncated Negative Binomial Regression Interpretation

A Zero-Truncated Negative Binomial model was fitted because hospital stay data showed overdispersion.

11.1 Age

$$\beta_{\text{age}} = 0.0816$$

$$IRR = e^{0.0816} \approx 1.085$$

Older patients tended to stay longer in the hospital.

11.2 Health Insurance

$$\beta_{\text{hmo}} = -0.4183$$

$$IRR = e^{-0.4183} \approx 0.658$$

Patients with HMO insurance had approximately 34.2% fewer expected hospital stay days.

11.3 Dispersion Parameter

$$\text{par0} = -1.0228$$

The significant dispersion parameter confirms overdispersion in the hospital stay data.

11.4 Important Interpretation

The contradictory age effect observed in the Zero-Truncated Poisson model suggests that overdispersion influenced coefficient estimation.

The Zero-Truncated Negative Binomial model is considered more reliable because it appropriately accounts for overdispersion.

12 Conclusion

The statistical analyses demonstrated significant associations among several categorical and count data variables.

Chi-Square tests and Fisher's Exact Tests identified significant relationships among disease status, wealth index, smoking, pollution exposure, and treatment effectiveness.

Regression analyses showed that demographic, academic, and behavioral variables significantly influence outcomes such as graduate admission, student absenteeism, award counts, fish-catching behavior, and hospital stay duration.

Poisson, Negative Binomial, Zero-Inflated, and Zero-Truncated models provided appropriate frameworks for analyzing count data under different distributional assumptions.

Overall, the findings highlight the importance of selecting suitable statistical models based on the characteristics of the data, including overdispersion and excess zero counts.